

Hands-on for 2D iterative reconstructions of simulated data with the CASToR platform

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0 Initializations

Just run it as it is. [1]

1 Definition and generation of the phantom

Background disk with smaller contrast disks

Generate both an ASCII file describing the phantom and a pixelized image of the phantom

Specify :

- the radius of the background disk in mm (radius), default = 100
- the emission density of the contrast disks relative to the background (sphereValues), default = 8 disks, from 4.0 to 0.0

Here is the phantom, called cylinder that we have generated :

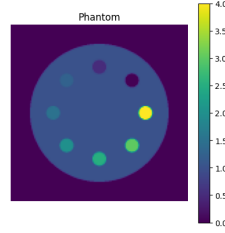


FIGURE 1.0.1 – Generation of the phantom

2 Simulation and reconstruction of a noise free sinogram

Left Image : Simulation of noise free data for phantom cylinder using scanner geometry PET_GE_SIGNAL_PET-MR

Right Image : Reconstruction of the noise free data with MLEM for 12 iterations and 14 subsets

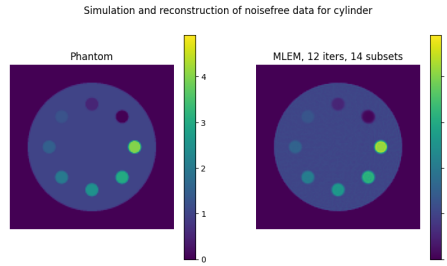


FIGURE 2.0.1 – Generation of the phantom

3 Notations & Introduction

Maximum Likelihood [2]

$$\hat{\theta}_{\text{ML}} = \arg \max_{\theta \in \Theta} L(\theta, m)$$

Maximum a posteriori

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta \in \Theta} (L(\theta, m) + \log p(\theta))$$

where we chose the Gibbs distribution as the a priori distribution, i.e. for the Markov random field prior :

$$p(\theta) = \frac{1}{Z} e^{-\beta U(\theta)},$$

with the Gibbs energy function

$$U(\theta) = \sum_{c \in \mathcal{C}} \psi(\theta),$$

in our case, we sum over its neighbors ($\sum_k$) and then sum over all pixels in the image ($\sum_j$), it gives

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta \in \Theta} \left(L(\theta, m) - \beta \sum_{j=1}^J \sum_{k \in N_j, k > j} \omega_{kj} \boxed{\psi(\theta_k, \theta_j)} \right)$$

where ψ is the potential function. In this Hands-on practice, we will choose among the following :

1. Quadratic

$$\psi_{\text{quad}}(\theta_k, \theta_j) = \frac{1}{2} \left(\frac{\theta_k - \theta_j}{\sigma} \right)^2$$

2. Huber

$$\psi_{\text{Huber}}(\theta_k, \theta_j) = \begin{cases} \delta |\theta_k - \theta_j| - \frac{\delta^2}{2} & \text{if } |\theta_k - \theta_j| > \delta, \\ \frac{1}{2} (\theta_k - \theta_j)^2 & \text{if } |\theta_k - \theta_j| \leq \delta. \end{cases}$$

3. Relative difference

$$\psi_{\text{RD}}(\theta_k, \theta_j) = \frac{(\theta_k - \theta_j)^2}{(\theta_k + \theta_j) + \gamma |\theta_k - \theta_j|}$$

Maximum likelihood - Expectation Maximization Algorithm

E-step

$$\text{Expectation } Q(\theta|\theta^{(p)}) = \mathbb{E} \left[L(\theta, n) \middle| m, \theta^{(p)} \right], \text{ with } L(\theta, n) := \log p(n|\theta)$$

M-step

$$\text{Maximization } \theta^{(p+1)} = \arg \max_{\theta \in \Theta} Q(\theta|\theta^{(p)})$$

where n is the complete data, which is actually unknown, while m represents the incomplete data.

For the ML-EM algorithm, we consider the complete data space n , assuming that n is known. By maximizing the new guess $\theta^{(p+1)}$ in each iteration (since we could compute $\mathbb{E}[n_{ij}|m_i, \theta^{(p)}]$ and $\mathbb{E}[n_i|m_i, \theta^{(p)}]$), the likelihood $L(\theta)$ increases monotonically, i.e., $L(\theta^{(p+1)}) \geq L(\theta^{(p)})$

In our case, the $L(\theta, m)$ for poisson distribution

$$L(\theta, m) = \sum_{i=1}^I \left(m_i \log \left(\sum_{j=1}^J a_{ij} \theta_j + b_i \right) - \left(\sum_{j=1}^J a_{ij} \theta_j + b_i \right) \right)$$

and assume the vector n as the complete data random vector

$$L(\theta, n) = \sum_{i=1}^I \left(\sum_{j=1}^J (n_{ij} \log(a_{ij} \theta_j) - a_{ij} \theta_j) + n_i \log(b_i) - b_i \right)$$

where

n_{ij} : The number of detected events originating from pixel j for LoR i .

n_i : The number of detected events from the background for LoR i .

Finally, we have

$$\theta^{(p+1)} = \arg \max_{\theta \in \Theta} \left(\sum_{i=1}^I \sum_{j=1}^J \left(m_i \frac{a_{ij} \theta_j^{(p)}}{\sum_{l=1}^J a_{il} \theta_l^{(p)} + b_i} \log(a_{ij} \theta_j) - a_{ij} \theta_j \right) \right)$$

OSEM acceleration [2] p160

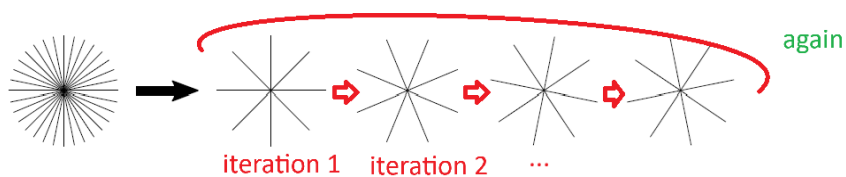


FIGURE 3.0.1 – Illustration | Acceleration of the ML-EM : Splitting in four subsets according to the azimuthal angle

4 Simulation and reconstruction of noisy data

We change the number of events to `eventsNb=10**6` to obtain a noisy image. We will use this value for the rest of the practical work.

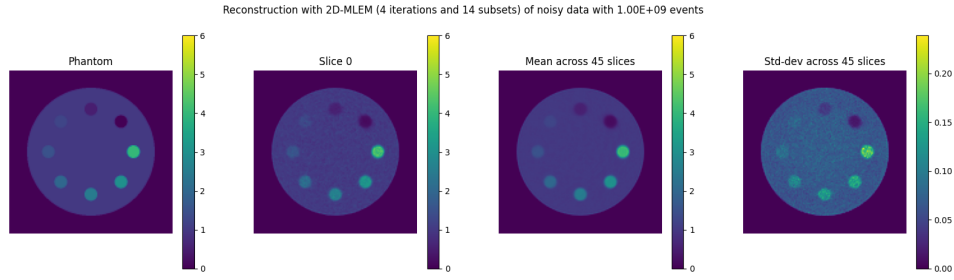


FIGURE 4.0.1 – Reconstruction with 2D-MLEM (4 iterations and 14 subsets) of noisy data with 1.00E+09 events.

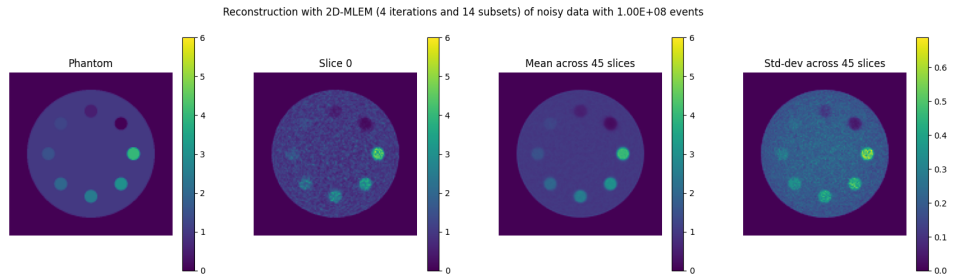


FIGURE 4.0.2 – Reconstruction with 2D-MLEM (4 iterations and 14 subsets) of noisy data with 1.00E+08 events.

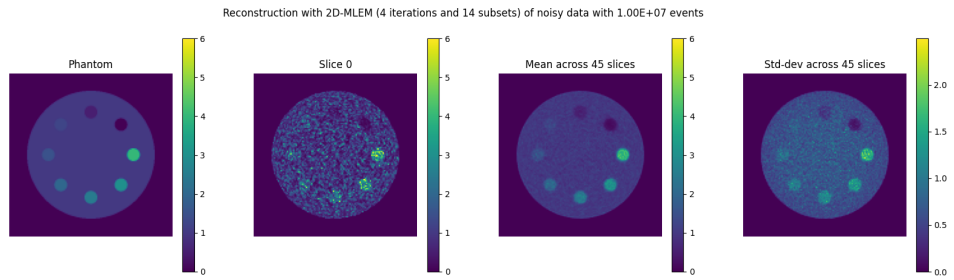


FIGURE 4.0.3 – Reconstruction with 2D-MLEM (4 iterations and 14 subsets) of noisy data with 1.00E+07 events.

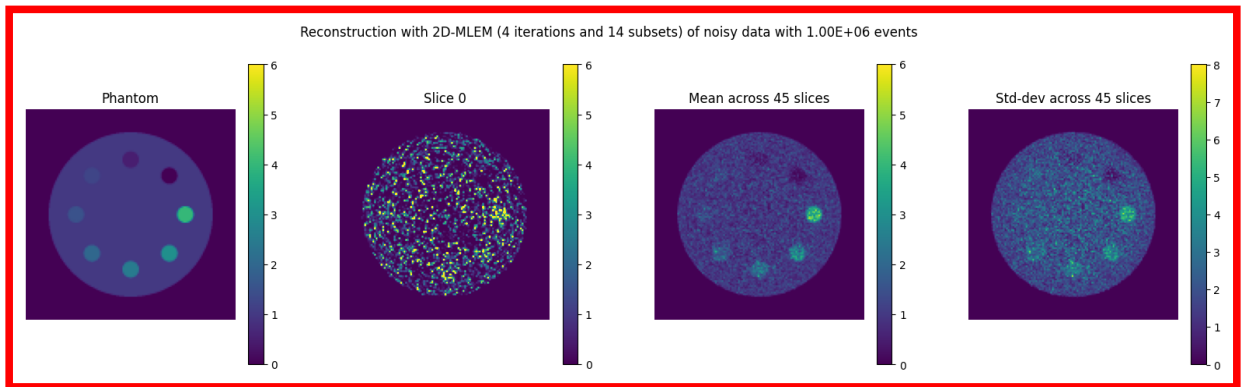


FIGURE 4.0.4 – Reconstruction with 2D-MLEM (4 iterations and 14 subsets) of noisy data with 1.00E+06 events.

5 Effect of the number of iterations for the ML-EM algorithm for noisy data

We plot the figure of the reconstruction of noisy data at different iteration counts :

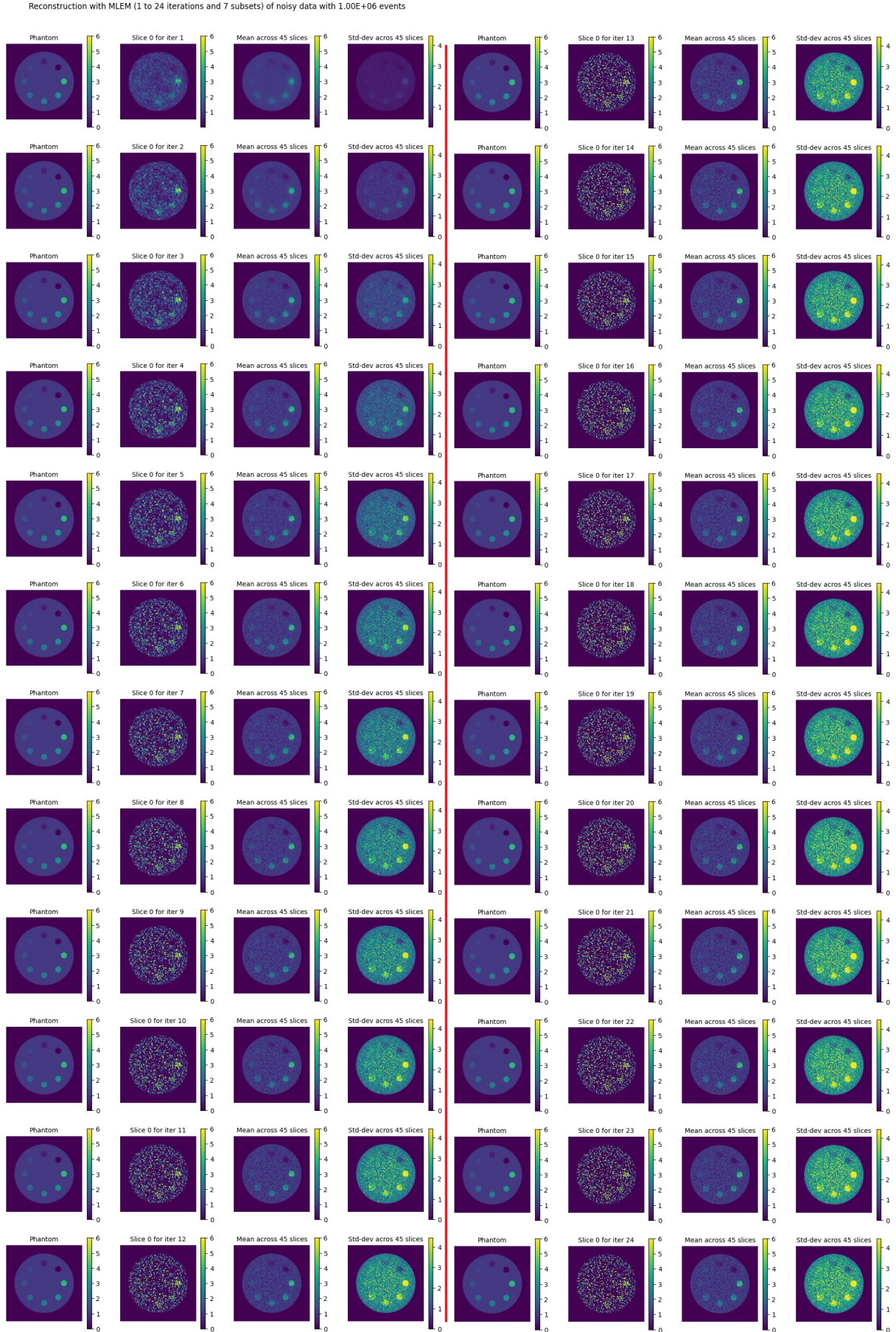


FIGURE 5.0.1 – Reconstruction with MLEM of the noisy sinogram with 1.00E+06 events

Also, for different contrast ratios (defined as $\text{contrast ratio} = \frac{\text{sphereValues}}{\text{backgroundValue}}$), we plotted the CRC (Contrast Recovery Coefficient, which reflects the recovery of the sphere's activity concentration after reconstruction, with a value of 1 being optimal) against Noise :

MLEM (1-24 iterations and 7 subsets) of noisy data with 1.00E+06 events

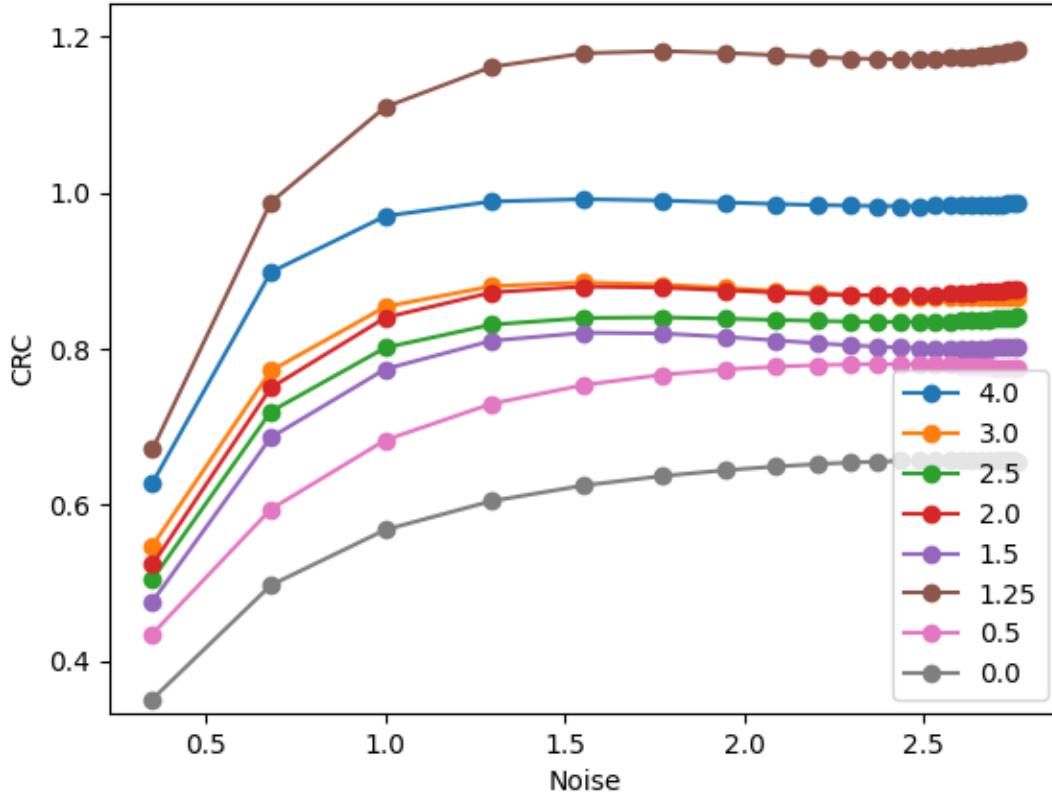


FIGURE 5.0.2 – MLEM (1-24 iterations and 7 subsets) of noisy data with 1.00E + 06 events

We notice that

1. The convergence speed is not the same for the different subsets ;
2. One of the subsets shows artifacts, causing the CRC to exceed the expected value ;
3. We also observe that the estimator becomes less accurate as the iterations progress.

6 Effect of the prior and the hyper-parameter value (beta) for the MAP algorithm for noisy data

For the MAP algorithm applied to noisy data, we generate two images for each of the different **potential functions** (quadratic, Huber, and RD) :

Left Image : Each row represents the effect of a different hyper-parameter value (β) on the reconstruction.

Right Image : The line plot shows the relationship between CRC (Contrast Recovery Coefficient) and Noise. Similarly, different lines represent different contrast ratios.

We notice that

1. The quadratic potential function results in an overly smoothed reconstruction image, leading to poorer performance, and therefore, we will not choose this function.
2. The Huber potential function smooths the reconstruction image while still allowing for significant changes ;
3. The relative difference potential function has a more open shape compared to the Huber function, which helps preserve contrast.

At the end of this subsection, we will compare and discuss the impact of the Huber and RD potential functions on the reconstruction results.

Here are the simulation results :

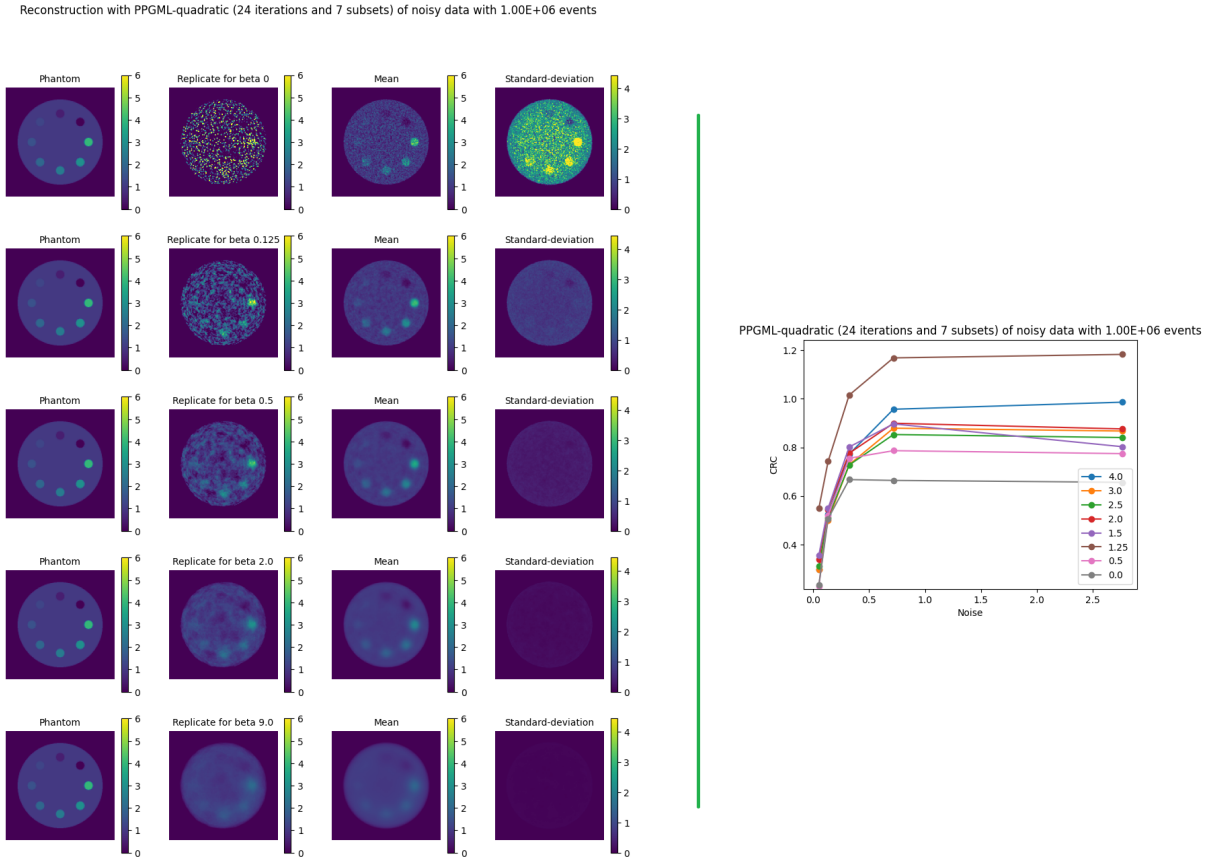
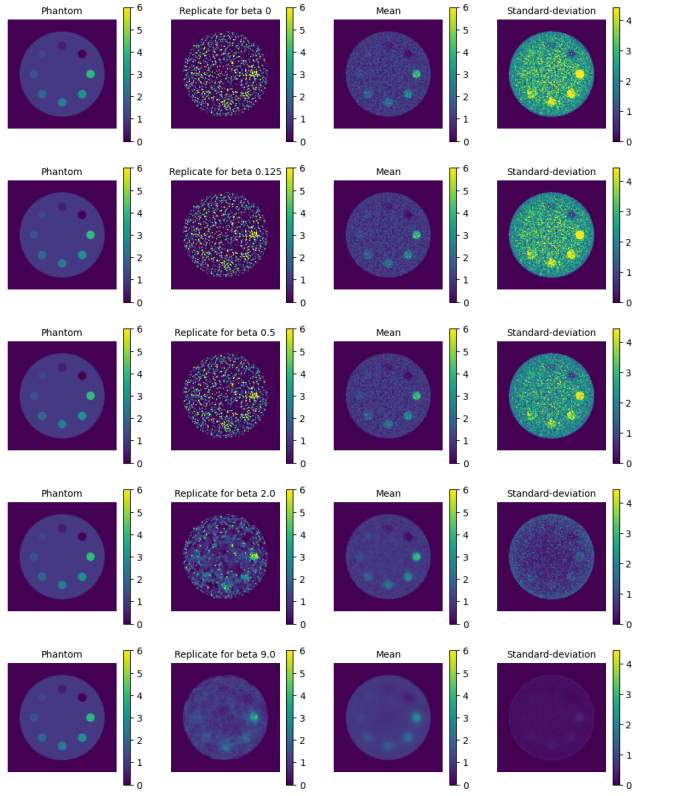


FIGURE 6.0.1 – Reconstruction with PPGML-**quadratic** (24 iterations and 7 subsets) of noisy data with 1.00E + 06 events

Reconstruction with PPGML-huber (24 iterations and 7 subsets) of noisy data with 1.00E+06 events



PPGML-huber (24 iterations and 7 subsets) of noisy data with 1.00E+06 events

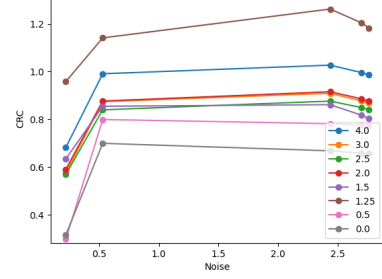
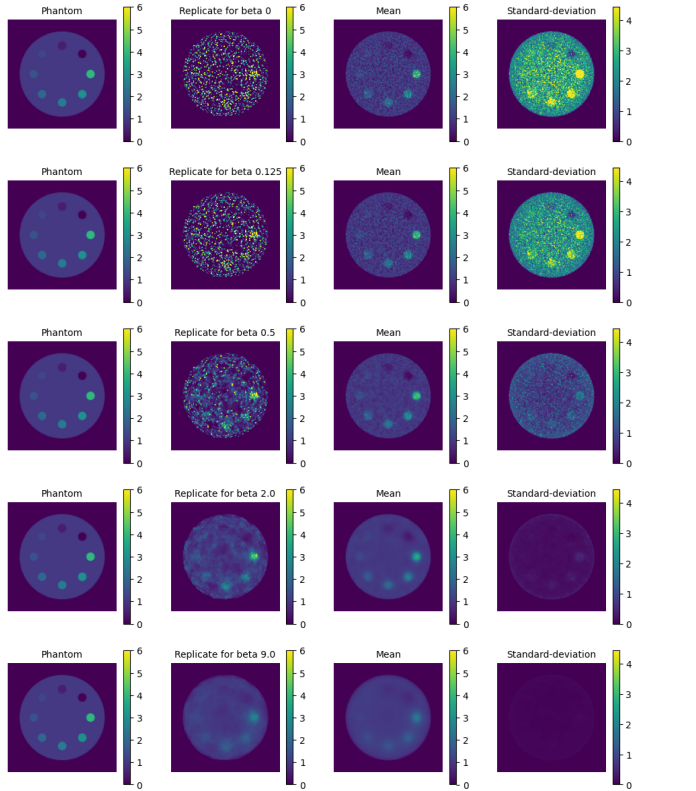


FIGURE 6.0.2 – Reconstruction with PPGML-huber (24 iterations and 7 subsets) of noisy data with 1.00E + 06 events

Reconstruction with PPGML-nuyts (24 iterations and 7 subsets) of noisy data with 1.00E+06 events



PPGML-nuyts (24 iterations and 7 subsets) of noisy data with 1.00E+06 events

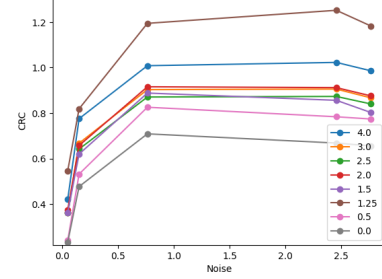


FIGURE 6.0.3 – Reconstruction with PPGML-nuyts (24 iterations and 7 subsets) of noisy data with 1.00E + 06 events

For the last two images (FIGURE 6.0.2 and FIGURE 6.0.3), we could place the case with the Huber potential function where $\beta = 2$ alongside the case with the Nuyts potential function where $\beta = 0.5$. There is not much difference between them :

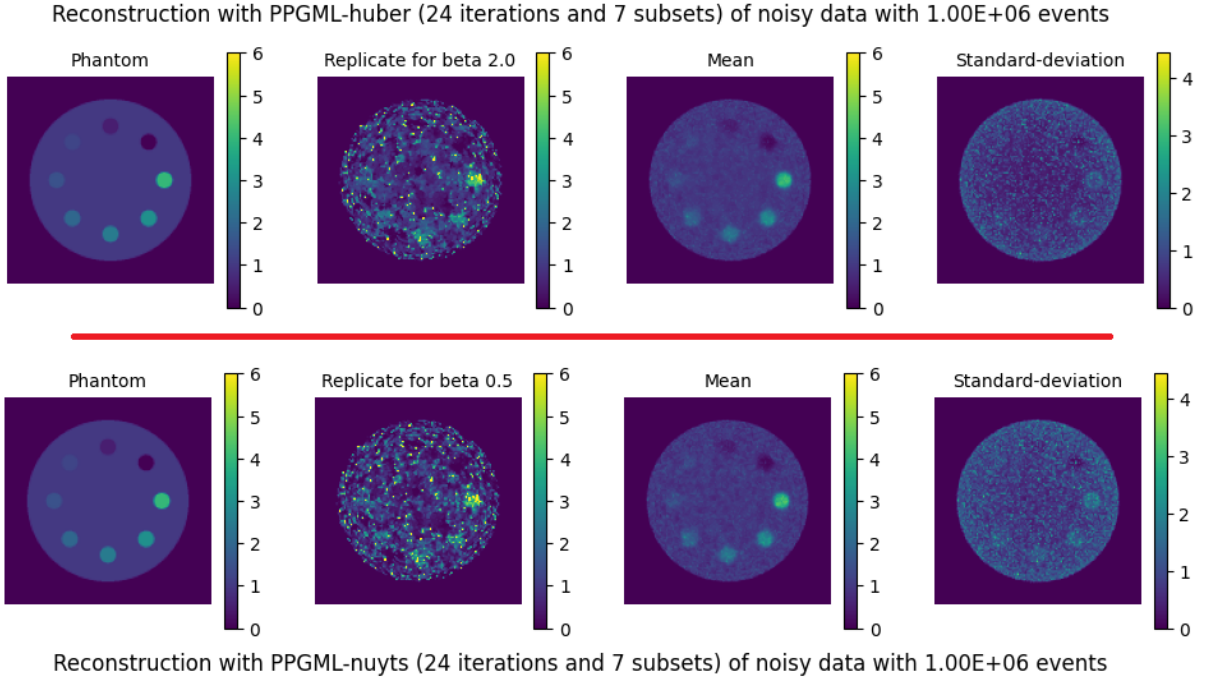


FIGURE 6.0.4 – Reconstruction with PPGML-**Huber** for $\beta = 2$ v.s. PPGML-**nuyts** for $\beta = 0.5$

Next, we could place the case with the Huber potential function where $\beta = 9$ together with the case where the Nuyts potential function has $\beta = 2$. The goal is to observe the contour. Personally, the contour for Nuyts with $\beta = 2$ is well-defined.

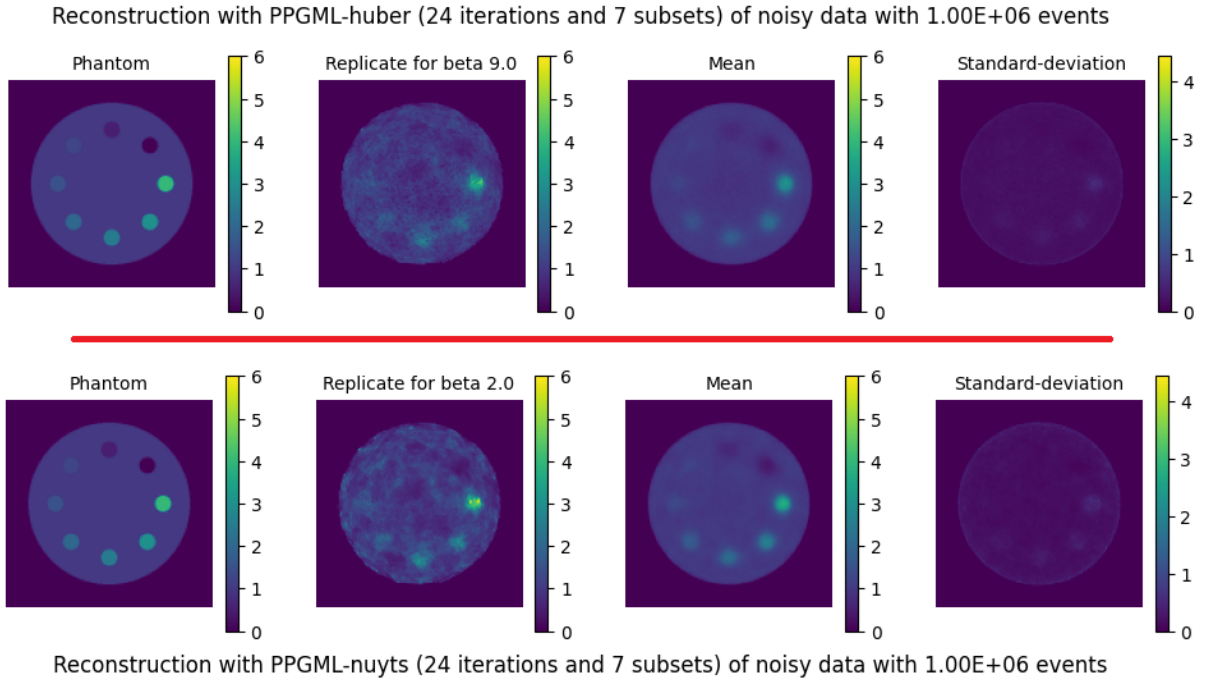


FIGURE 6.0.5 – Reconstruction with PPGML-**Huber** for $\beta = 9$ v.s. PPGML-**nuyts** for $\beta = 2$

Ultimately, this is about visualization.

Considering the standard deviation, we should analyze the FIGURE 6.0.6 : typically, for a given noise level, we draw a vertical line, and the method with the curve closest to 1 represents the best performance. In our case, all subplots appear similar, and for noise = 0.5, for example, the three PPGML methods (quadratic, Huber, and RD) perform quite similarly :

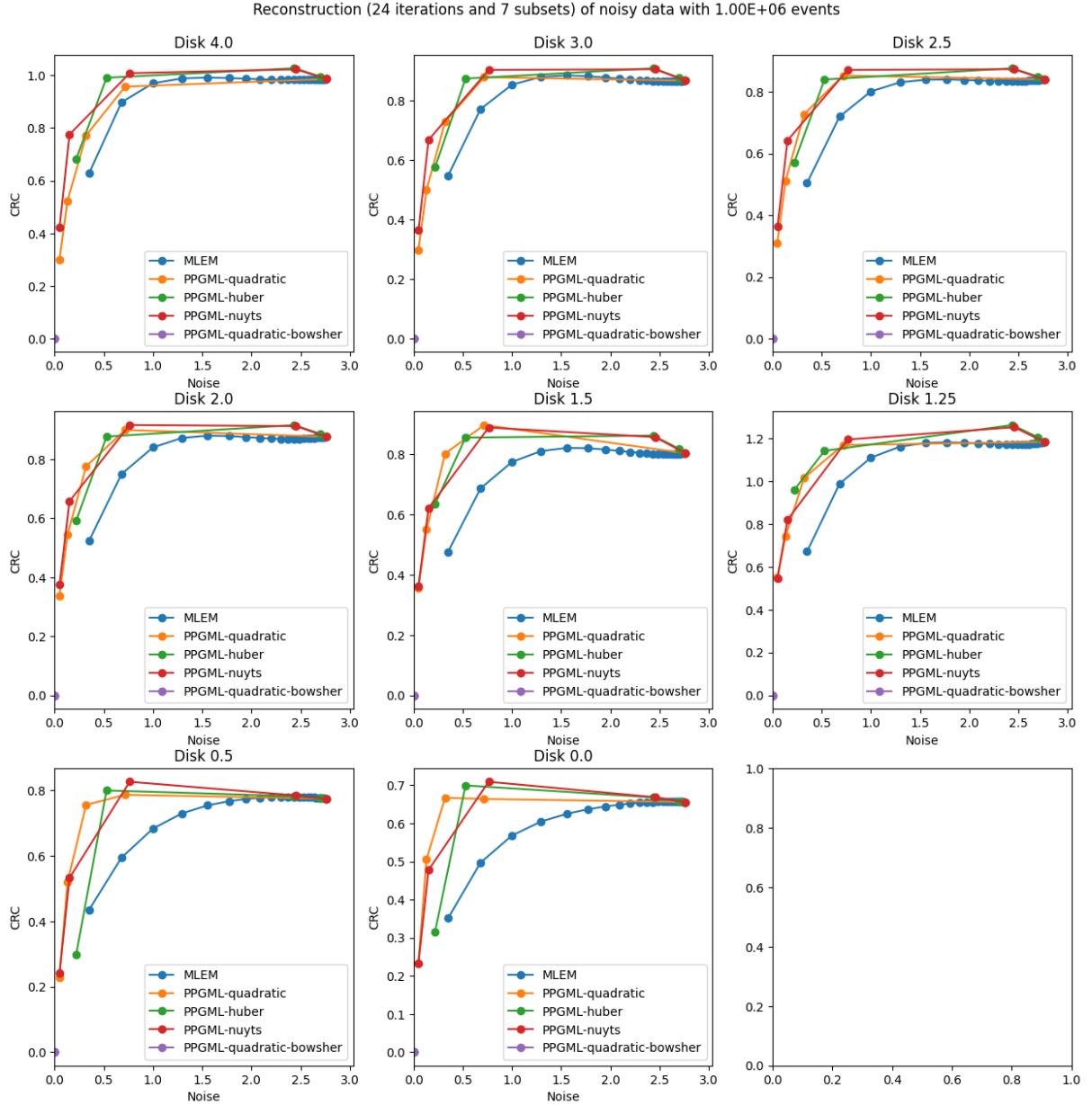


FIGURE 6.0.6 – CRC against Noise for different contrast ratios across various reconstruction methods

7 Conclusion

During this Hands-on

1. We perform the simulation and reconstruction of noisy data, selecting the number of events as `eventsNb=10**6` for the remainder of the hands-on session.
2. We analyze the effect of the number of iterations on the ML-EM algorithm for noisy data.
3. Finally, we investigate the effect of the prior and the hyper-parameter value (β) in the MAP algorithm for noisy data.

Références

- [1] C. Comtat. *Hands-on for 2D iterative reconstructions of simulated data with the CASToR platform*. Université Paris-Saclay, 2023-2024.
- [2] C. Comtat. *M2 CONTROL, SIGNAL AND IMAGE PROCESSING UE Medical Imaging Positron Emission Tomography*. Université Paris-Saclay, 2023-2024.